

# A First Look at Your Project's Size

CS 5388 / CS 4390

**Team:** \_\_\_\_ Team 6 – The Gentlemen \_\_\_\_ **Project:** Universal Automotive Management App \_\_\_\_

Work through every step together as a team. Have your Sprint 1 charter open (scope boundary and stakeholder map) and your interview notes. Everything you write here becomes the starting point for today's user stories and for the two estimates in Part 2.

## Step 1. List what Release 1 must do (as a team)

Start from the “in scope” items in your charter and the strongest needs from your interviews. Write 5 to 8 capabilities as what the system must do, not how. Then mark the hidden work each one will need using the key below.

**Hidden work key:** T = Testing S = Security or privacy I = Integration with other systems D = Data (collection, cleaning, migration) H = Hosting and deployment U = Usability and accessibility O = Documentation or training

| #   | Capability (what, not how)                                                                        | Evidence (charter or interview)      | Hidden work       |
|-----|---------------------------------------------------------------------------------------------------|--------------------------------------|-------------------|
| Ex. | <i>A student can buy a semester parking permit online.</i>                                        | <i>Interview 4, Parking Services</i> | <i>T, S, I, H</i> |
| 1   | It stores basic vehicle information                                                               | Charter                              | T, S, D           |
| 2   | Gives recommended maintenance on the vehicle                                                      | Interview 1, Charter                 | T, D, U           |
| 3   | Keeps track of miles user travelled                                                               | Charter                              | T, S, I, D        |
| 4   | Gives maintenance reminders for routine services                                                  | Interview 6, Charter                 | T, D, U, O        |
| 5   | Keeps maintenance and repair history for each vehicle                                             | Interview 8, Charter                 | T, D              |
| 6   | Allows user to record maintenance expenses, including service performed, date, mileage, and cost. | Interview 7, Charter                 | T, D, U           |
| 7   | Provides a summary of maintenance costs for each vehicle                                          | Interview 11, Charter                | T, D              |
| 8   |                                                                                                   |                                      |                   |

*The example row is from the fictional Campus Parking System used in lecture.*

## Step 2. Take the outside view: your own experience (as a team)

As a team, pick the most similar thing any of you has built before: a class project, internship feature, hackathon app, or side project.

| Question     | Answer                                                                                                                                                                                                |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| What was it? | Database Systems Project: a web app where you could manage and make appointments for any of your pets at the vet. Could choose any vet and vets had access to all medical records for their patients. |

| Question                          | Answer                                                                                                                                   |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| How long was it expected to take? | Was a semester long project but started it around halfway so about 8 weeks.                                                              |
| How long did it take?             | Took the 8 weeks but some features took longer to implement than expected.                                                               |
| What took longer than expected?   | There were a few errors coming up here and there. Some things that were not working within the database took a bit longer than expected. |

### Step 3. Take the outside view: similar products *(as a team, web search)*

Divide the searching, then decide together. Find three real products that solve a problem close to yours. Include at least one small product (a startup or an app from a small team) and at least one big product (a large company platform or a government system). For each, find out how long it took to reach its first public release, what went well, and what went wrong.

#### How to find the timeline and the story

1. Launch announcements, company blogs, and founder interviews often say when building started and when the product first went public.
2. Press coverage, app store version histories, and the Internet Archive's Wayback Machine help confirm first release dates.
3. Postmortems, conference talks, and, for government systems, GAO and Inspector General reports explain what worked and what did not.
4. Measure from the start of development to the first public release, and write down what that first release included. A limited first version and a full launch are different numbers.
5. Every date and figure needs a source you can link. If you cannot find one, write "unknown." AI tools may suggest products to look at, but they are not a source.

#### Example: a completed entry for a parking product

*ParkEasy is fictional. It shows the level of detail to aim for, not real data.*

| Field                           | Entry                                                                                                                                                                    |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Product and company             | ParkEasy, a four-person startup                                                                                                                                          |
| Why it is similar               | Drivers find and pay for parking from a phone, like our Campus Parking System.                                                                                           |
| Development started             | January 2019 (founder blog post)                                                                                                                                         |
| First public release            | October 2019, in one city only (local news launch story)                                                                                                                 |
| Time to first release           | About 9 months                                                                                                                                                           |
| What the first release included | A map of open lots and pay-by-phone. No reservations and no monthly permits.                                                                                             |
| Team size                       | Three engineers and one designer                                                                                                                                         |
| What worked                     | Launching in one city first. Partnering with a single lot operator to get reliable availability data.                                                                    |
| What did not work               | Their own parking sensors gave bad availability data, so they switched to the operator's feed, adding about 3 months. Payment integration took twice as long as planned. |
| Sources                         | Founder blog post (link); local news launch story (link)                                                                                                                 |

*What this example tells a team building a parking app: plan for data and payment integration to take the longest, and scope the first release to one location.*

### Step 3 worksheet: similar products

|                                   | Product A (small)                                                                                                                                                                                                                                                                                                                   | Product B (big)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Product C (your choice)                                                                                          |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Product and company               | AUTOsist, founded by Zorrane Abdeali.                                                                                                                                                                                                                                                                                               | myCARFAX, now CARFAX Car Care, developed by CARFAX.                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Drivvo, originally CTN Car, a small independent Brazilian team                                                   |
| Why it is similar to your project | Tracks vehicle maintenance, service records, fuel, and expenses.                                                                                                                                                                                                                                                                    | Provides service history, maintenance reminders, recall information, and vehicle information.                                                                                                                                                                                                                                                                                                                                                                                                                        | Tracks vehicle fuel, expenses, maintenance and service history                                                   |
| Development started               | Abdeali came up with and started working on it in 2014                                                                                                                                                                                                                                                                              | Around 2011                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Unknown                                                                                                          |
| First public release              | 2014/Early 2015, initially shared with friends and car enthusiasts.                                                                                                                                                                                                                                                                 | August 22, 2012 on the App Store                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Platform launched in 2012                                                                                        |
| Time to first release             | Exact time unknown but it appears to be around a year.                                                                                                                                                                                                                                                                              | Exact time also unknown but appears to be 1 year – 1 ½ years                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Unknown                                                                                                          |
| What the first release included   | It was a consumer-focused app for storing and sharing vehicle maintenance information.                                                                                                                                                                                                                                              | Vehicle service-history tracking, automatic service alerts, safety-recall information, and access through iPhone, Android, and the web.                                                                                                                                                                                                                                                                                                                                                                              | A simple management app the focused on tracking the vehicle info                                                 |
| Team size, if known               | One person at launch, Abdeali                                                                                                                                                                                                                                                                                                       | Unknown                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Unknown, described only as a small independent team                                                              |
| What worked                       | He addressed a real problem he faced, not having a place to view all his personal vehicle maintenance history.                                                                                                                                                                                                                      | They partnered with thousands of existing service shops, so history filled in automatically instead of having to manually fill it out                                                                                                                                                                                                                                                                                                                                                                                | It was simple which allowed users to get the hang of it easily                                                   |
| What did not work                 | Business users wanted to use the app for fleets which the app initially didn't accommodate. He listened and shifted it into a commercial fleet management tool in December 2016                                                                                                                                                     | Unknown, none found in available sources                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | It was only limited to certain languages and currencies                                                          |
| Sources (links)                   | Founder Interview ( <a href="https://www.ccjdigital.com/business/article/14934261/autosist-app-simplifies-maintenance-recordkeeping">https://www.ccjdigital.com/business/article/14934261/autosist-app-simplifies-maintenance-recordkeeping</a> )<br>Company Website: ( <a href="https://autosist.com/">https://autosist.com/</a> ) | Carfax CIO interview ( <a href="https://www.cio.com/article/291032/mobile-don-t-let-mobile-development-ruin-your-life.html">https://www.cio.com/article/291032/mobile-don-t-let-mobile-development-ruin-your-life.html</a> )<br>Carfax press release ( <a href="https://www.vehicleservicepros.com/shop-operations/service-repair/news/10965058/carfax-unveils-new-car-care-mobile-app">https://www.vehicleservicepros.com/shop-operations/service-repair/news/10965058/carfax-unveils-new-car-care-mobile-app</a> ) | Drivvo Website ( <a href="https://www.drivvo.com/en/personal-use/">https://www.drivvo.com/en/personal-use/</a> ) |

#### Step 4. Compare *(as a team)*

1. Look at the hidden-work column in Step 1. Which letter appears most often? Did any of your similar products struggle with that same kind of work?

T and D are the ones that appear most often in our work. Our similar products also dealt with the data related challenges. AUSTOsist was created with the intention of making vehicle maintenance info organized and CARFAX benefited from partnering with some service shops so that the vehicle history could automatically be added instead of having to do that manually. This backs up our belief of the importance of properly collecting and storing vehicle data in our project.

2. How long did your similar products take to reach a first release, and with how many people? What does that suggest about a first release of your project?

From what we saw AUTOsist took around a year to reach its first release with only 1 person while CARFAX took somewhere around a year and a half but we do not know what the team size was just like Drivvo. After researching these products, the lesson we took out of it was that we should focus on getting the most important features correct instead of trying to get everything done at once. We believe our project could take around 10-15 months to get it done!

3. Name one practice from the “what worked” rows your team will copy, and one mistake from the “what did not work” rows you will avoid.

For these similar products’ first releases, they stayed narrow with the functionality, they only did what they needed to which is what we will copy. For the mistake we will avoid what Drivvo did and make sure our app can be used by people internationally.

**Our comparables suggest a first release would take** \_\_\_10\_\_\_ to \_\_\_15\_\_\_ months with a team of \_\_\_4\_\_\_.

*Keep this sheet. You will use Step 1 to write user stories, and Steps 2 and 3 give you comparators for analogous estimation in Part 2.*